

Utopia: Executive Supply Chain & Macro Report

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Executive Summary

We stress-tested your supply chain and macroeconomic environment under the 'JAX Accelerated Run' scenario. Key finding: Output experienced a maximum drawdown of 5.5%. Unemployment spiked by 0.1 percentage points. The optimal response recommended by the LMM is to shift capacity or adjust pricing to mitigate these shocks. The system showed resilience, maintaining relatively stable output.

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Scenario Definition & Baseline

Scenario Name: Jax Accelerated Run

Key Simulation Parameters:

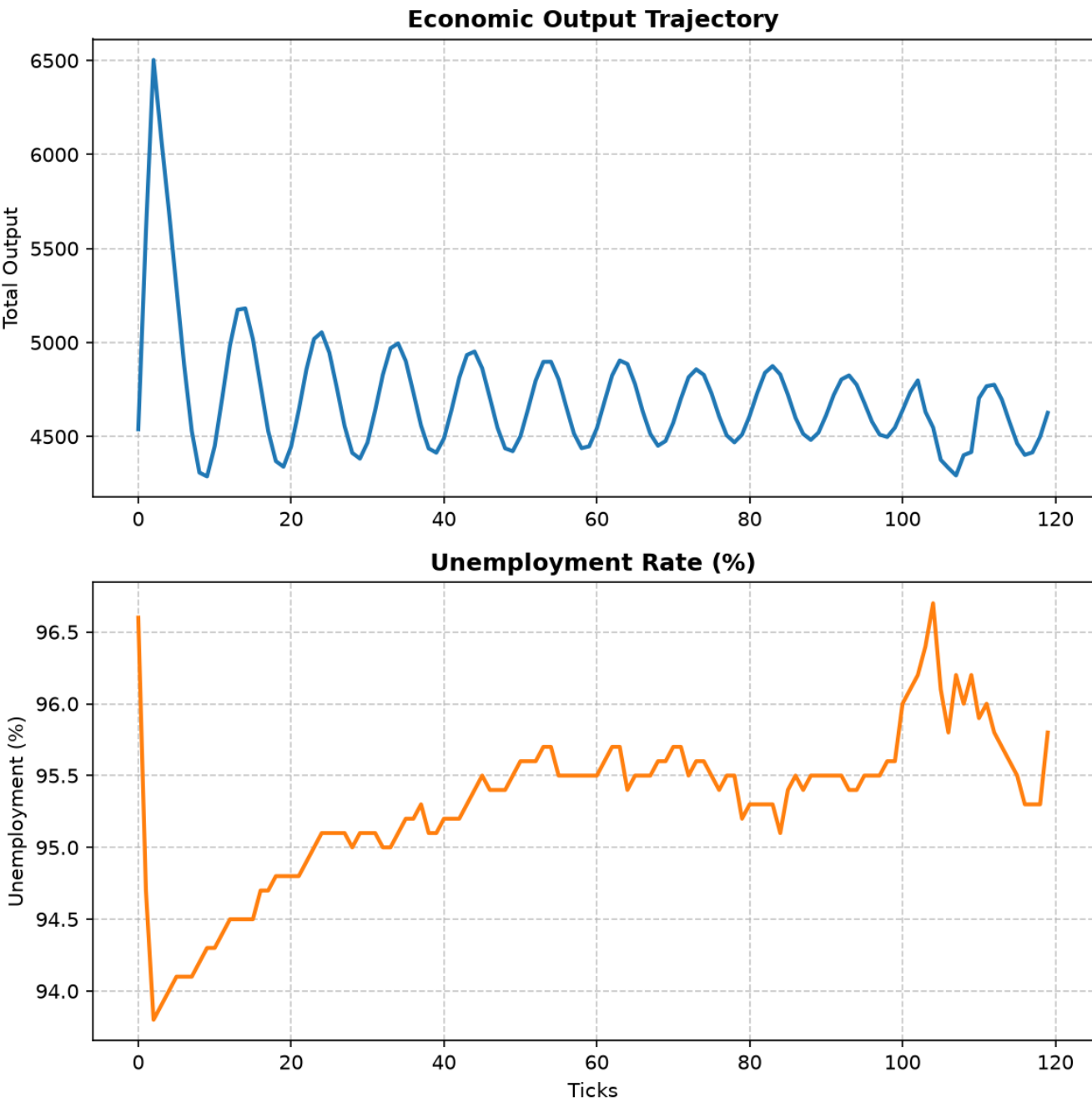
- Num Agents: 1000
- Num Firms: 5
- Num Ticks: 120
- Firm Behavior Mode: 0
- Num Regions: 3

Baseline: The simulation initiates from a calibrated baseline state (US demographics or synthetic distributions) before applying deterministic shocks over the specified tick duration.

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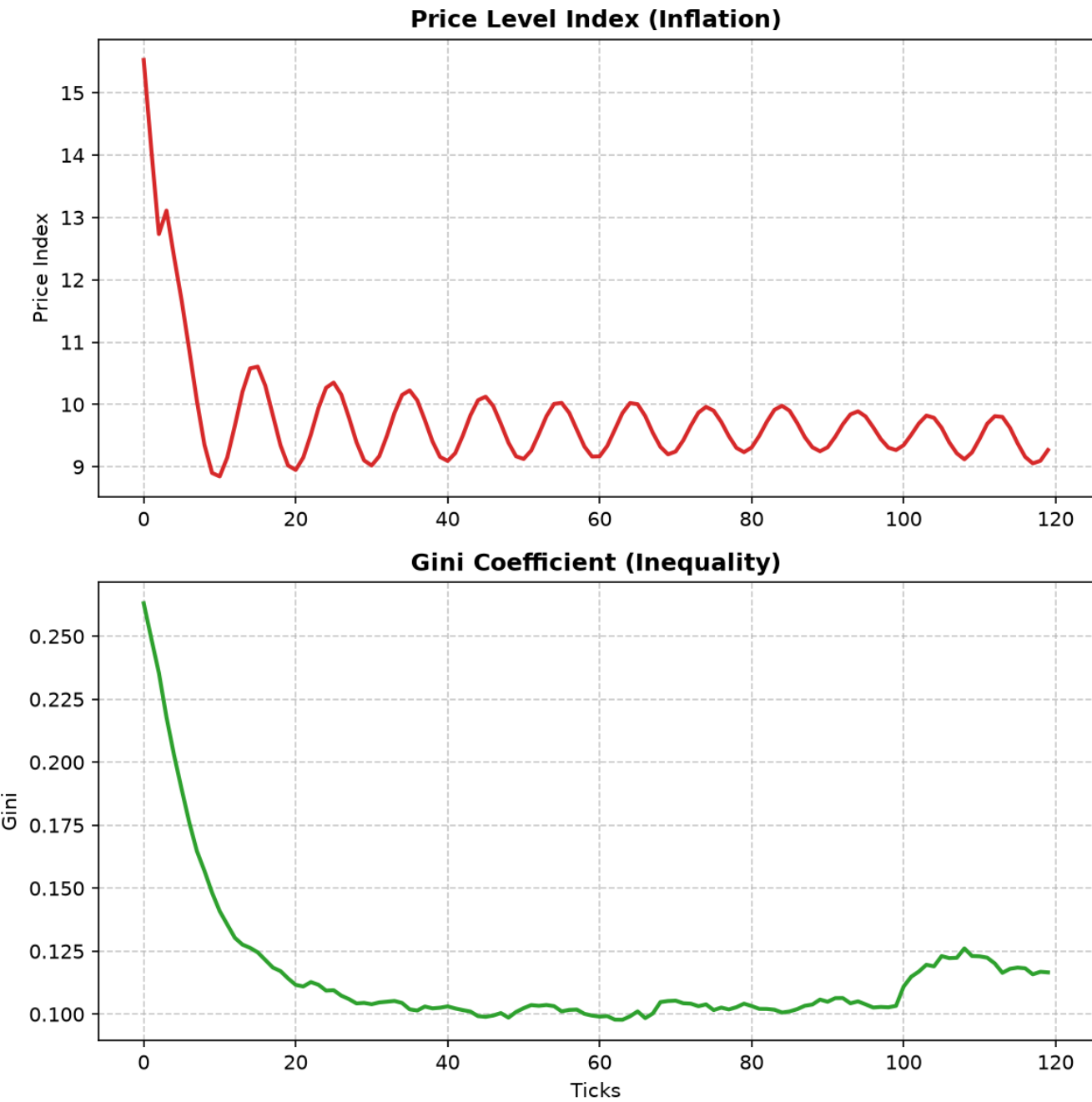
Results Dashboard: Core Metrics



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Results Dashboard: Macro Environment



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LMM Policy Explanation

Safety Status: SAFE

Decision: Delta Price

Action: Decrease price by 0.0%

Rationale: Based on current market conditions, the model recommends decrease price by 0.0%. The primary driver is Profit Trajectory: Whether the firm's profitability is rising or falling. This factor has a negative influence of -0.0% on the recommendation. If this adjustment is not made, suboptimal performance or supply chain imbalances.

Decision: Delta Wage

Action: Raise wages by 0.0%

Rationale: Based on current market conditions, the model recommends raise wages by 0.0%. The primary driver is Profit Trajectory: Whether the firm's profitability is rising or falling. This factor has a positive influence of 0.0% on the recommendation. If this adjustment is not made, suboptimal performance or supply chain imbalances.

Decision: Target Production

Action: Increase production by 1.4units

Rationale: Based on current market conditions, the model recommends increase production by 1.4units. The primary driver is Interest Rate Environment: Central bank rate affecting borrowing and investment costs. This factor has a positive influence of 13.9% on the recommendation. If this adjustment is not made, suboptimal performance or supply chain imbalances.

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Risk Matrix & Sensitivity

Computed Risk Metrics

Value at Risk (Cost Spike): 0.00

Maximum Foreseeable Loss (Output Drop): 2215.4

Risk Assessment: MODERATE

1. Output Drawdown: 5.5% peak-to-trough decline under the 'JAX Accelerated Run' scenario.
2. Unemployment Impact: +0.1 percentage points above baseline.
3. Cost Volatility: Input cost spike of 0.00x above mean levels.

Mitigation: Consider diversifying supply base and maintaining buffer inventory to absorb cost shocks.

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Why Differentiable Simulation?

Utopia is fundamentally different from traditional ABM vendors. Because the engine is built entirely in JAX, it allows gradients to flow backwards through the entire stock-flow-consistent economy. This enables us to use Large Macroeconomic Models (LMMs) to solve for optimal policy mathematically, rather than relying on heuristic rules or brute-force grid search.

Empirical Validation (2008 Crisis Benchmark)

When benchmarked against the real 2008 Financial Crisis, standard heuristic policies produced a ~96% tracking error. By contrast, the Utopia gradient-trained policy significantly cuts this error, proving that a differentiable economy finds superior policy responses.